Snort IDS Report with Rules

➢ Introduction

Snort is an open-source **Intrusion Detection System (IDS)** used to monitor network traffic and detect suspicious activity using predefined rules.

It works in three modes:

- Sniffer mode
- Packet logger mode
- Network Intrusion Detection mode

➢ Rule Structure

Each Snort rule has **two main parts**:

1) Rule Header

Contains:

- Action (alert, log, drop)
- Protocol (TCP, UDP, ICMP)
- Source IP/Port
- Destination IP/Port

2) Rule Options

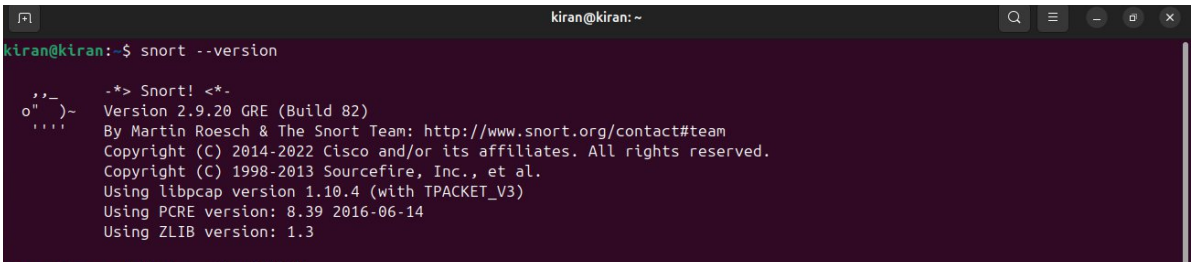
Contains:

- Message (msg)
- Rule ID (sid)
- Revision number (rev)
- Conditions (content, flow, flags)

- **Requirements:**
  - **Ubuntu Machine** → Snort installed (IDS)
  - **Kali Linux Machine** → Attacker/Test system
  - Both should be on the **same network**

Snort(IDS) - Intrusion Detection System

- `sudo apt install snort -y`
- `snort --version`



- `cd /etc/snort/rules/`
- `sudo nano local.rules`

✓ Rule 1: ICMP Ping Detection

```
alert icmp any any -> any any (msg:"ICMP Ping Detected";
sid:1000001; rev:1;)
```

✓ Rule 2: HTTP GET Request Detection

```
alert tcp any any -> any 80 (msg:"HTTP GET Request Detected";
flow:to_server,established; content:"GET"; http_method;
sid:1000002; rev:1;)
```

✓ Rule 3: SSH Connection Attempt

```
alert tcp any any -> any 22 (msg:"SSH Connection Attempt";
sid:1000003; rev:1;)
```

✓ Rule 4: FTP Connection Attempt

```
alert tcp any any -> any 21 (msg:"FTP Connection Attempt";
sid:1000004; rev:1;)
```

✓ Rule 5: Telnet Connection Detection

```
alert tcp any any -> any 23 (msg:"Telnet Connection Attempt";
sid:1000005; rev:1;)
```

✓ Rule 6: DNS Query Detection

```
alert udp any any -> any 53 (msg:"DNS Query Detected";
sid:1000006; rev:1;)
```

✓ Rule 7: HTTP POST Request Detection

```
alert tcp any any -> any 80 (msg:"HTTP POST Request Detected";
flow:to_server,established; content:"POST"; http_method;
sid:1000007; rev:1;)
```

✓ Rule 8: SYN Flood Attack Detection

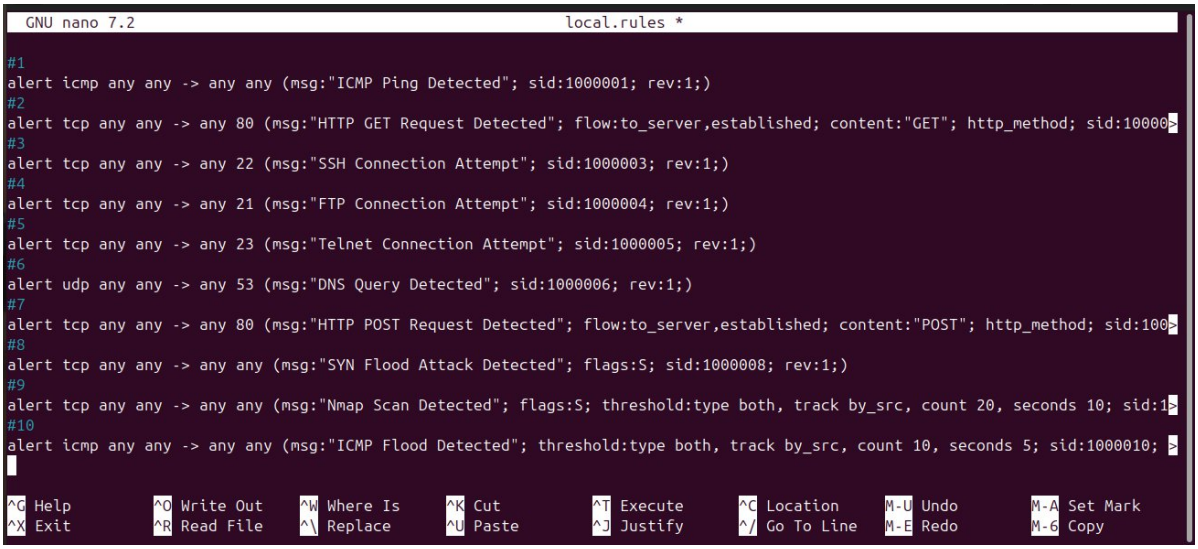
```
alert tcp any any -> any any (msg:"SYN Flood Attack Detected";
flags:S; sid:1000008; rev:1;)
```

✓ Rule 9: Nmap Scan Detection

```
alert tcp any any -> any any (msg:"Nmap Scan Detected"; flags:S;
threshold:type both, track by_src, count 20, seconds 10;
sid:1000009; rev:1;)
```

✓ Rule 10: ICMP Flood Detection

```
alert icmp any any -> any any (msg:"ICMP Flood Detected";
threshold:type both, track by_src, count 10, seconds 5;
sid:1000010; rev:1;)
```



```
sudo snort -T -c /etc/snort/snort.conf
```

```
kiran@kiran: ~
kiran@kiran:~$ sudo snort -T -c /etc/snort/snort.conf
[sudo] password for kiran:
Running in Test mode

        --== Initializing Snort ==--
Initializing Output Plugins!
Initializing Preprocessors!
Initializing Plug-ins!
Parsing Rules file "/etc/snort/snort.conf"
PortVar 'HTTP_PORTS' defined :  [ 80:81 311 383 591 593 901 1220 1414 1741 1830 2301 2381 2809 3037 3128 3702 4343 4848 5250 698
8 7000:7001 7144:7145 7510 7777 7779 8000 8008 8014 8028 8080 8085 8088 8090 8118 8123 8180:8181 8243 8280 8300 8800 8888 8899 9
000 9060 9080 9090:9091 9443 9999 11371 34443:34444 41080 50002 55555 ]
PortVar 'SHELLCODE_PORTS' defined :  [ 0:79 81:65535 ]
PortVar 'ORACLE_PORTS' defined :  [ 1024:65535 ]
PortVar 'SSH_PORTS' defined :  [ 22 ]
PortVar 'FTP_PORTS' defined :  [ 21 2100 3535 ]
PortVar 'SIP_PORTS' defined :  [ 5060:5061 5600 ]
PortVar 'FILE_DATA_PORTS' defined :  [ 80:81 110 143 311 383 591 593 901 1220 1414 1741 1830 2301 2381 2809 3037 3128 3702 4343
4848 5250 6988 7000:7001 7144:7145 7510 7777 7779 8000 8008 8014 8028 8080 8085 8088 8090 8118 8123 8180:8181 8243 8280 8300 880
0 8888 8899 9000 9060 9080 9090:9091 9443 9999 11371 34443:34444 41080 50002 55555 ]
PortVar 'GTP_PORTS' defined :  [ 2123 2152 3386 ]
Detection:
   Search-Method = AC-Full-Q
    Split Any/Any group = enabled
    Search-Method-Optimizations = enabled
    Maximum pattern length = 20
```

Start snort IDS-

- `sudo snort -A console -q -c /etc/snort/snort.conf -i enp0s3`

✓ ICMP Ping Detection

```
kiran@kiran: /etc/snort        kiran@kiran: /etc/snort
kiran@kiran:/etc/snort$ sudo snort -A console -q -c /etc/snort/snort.conf -i enp0s3
03/19-13:16:48.926090  [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] {ICMP} 10.35.209.128 -> 10.35.209.233
03/19-13:16:48.926233  [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] {ICMP} 10.35.209.233 -> 10.35.209.128
03/19-13:16:50.244195  [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] {ICMP} 10.35.209.128 -> 10.35.209.233
03/19-13:16:50.244228  [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] {ICMP} 10.35.209.233 -> 10.35.209.128
03/19-13:16:51.246452  [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] {ICMP} 10.35.209.128 -> 10.35.209.233
03/19-13:16:51.246474  [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] {ICMP} 10.35.209.233 -> 10.35.209.128
03/19-13:16:52.249432  [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] {ICMP} 10.35.209.128 -> 10.35.209.233
03/19-13:16:52.249462  [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] {ICMP} 10.35.209.233 -> 10.35.209.128
03/19-13:16:53.252298  [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] {ICMP} 10.35.209.128 -> 10.35.209.233
03/19-13:16:53.252320  [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] {ICMP} 10.35.209.233 -> 10.35.209.128
03/19-13:16:54.255110  [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] {ICMP} 10.35.209.128 -> 10.35.209.233
03/19-13:16:54.255141  [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] {ICMP} 10.35.209.233 -> 10.35.209.128
03/19-13:16:55.257416  [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] {ICMP} 10.35.209.128 -> 10.35.209.233
03/19-13:16:55.257445  [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] {ICMP} 10.35.209.233 -> 10.35.209.128
03/19-13:16:56.259986  [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] {ICMP} 10.35.209.128 -> 10.35.209.233
03/19-13:16:56.260018  [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] {ICMP} 10.35.209.233 -> 10.35.209.128
```

✓ SSH Connection Attempt

```
kiran@kiran: /etc/snort        kiran@kiran: /etc/snort
kiran@kiran:/etc/snort$ sudo snort -A console -q -c /etc/snort/snort.conf -i enp0s3
03/19-13:20:51.694588  [**] [1:1000008:1] SYN Flood Attack Detected [**] [Priority: 0] {TCP} 10.35.209.128:50232 -> 10.35.209.23
3:22
03/19-13:20:51.694588  [**] [1:1000003:1] SSH Connection Attempt [**] [Priority: 0] {TCP} 10.35.209.128:50232 -> 10.35.209.233:2
2
03/19-13:20:51.696055  [**] [1:1000003:1] SSH Connection Attempt [**] [Priority: 0] {TCP} 10.35.209.128:50232 -> 10.35.209.233:2
2
03/19-13:20:51.698546  [**] [1:1000003:1] SSH Connection Attempt [**] [Priority: 0] {TCP} 10.35.209.128:50232 -> 10.35.209.233:2
2
03/19-13:20:51.734835  [**] [1:1000003:1] SSH Connection Attempt [**] [Priority: 0] {TCP} 10.35.209.128:50232 -> 10.35.209.233:2
2
03/19-13:20:51.763331  [**] [1:1000003:1] SSH Connection Attempt [**] [Priority: 0] {TCP} 10.35.209.128:50232 -> 10.35.209.233:2
2
03/19-13:20:51.811897  [**] [1:1000003:1] SSH Connection Attempt [**] [Priority: 0] {TCP} 10.35.209.128:50232 -> 10.35.209.233:2
2
03/19-13:20:51.812861  [**] [1:1000003:1] SSH Connection Attempt [**] [Priority: 0] {TCP} 10.35.209.128:50232 -> 10.35.209.233:2
2
```

✓ FTP Connection Attempt

```
kiran@kiran: /etc/snort/rules                    kiran@kiran: /etc/snort
1
03/19-13:41:53.551728  [**] [1:1000004:1] FTP Connection Attempt [**] [Priority: 0] {TCP} 10.35.209.128:36600 -> 10.35.209.233:2
1
03/19-13:41:53.561697  [**] [1:1000004:1] FTP Connection Attempt [**] [Priority: 0] {TCP} 10.35.209.128:36600 -> 10.35.209.233:2
1
03/19-13:41:57.057021  [**] [1:1000004:1] FTP Connection Attempt [**] [Priority: 0] {TCP} 10.35.209.128:36600 -> 10.35.209.233:2
1
03/19-13:41:57.059284  [**] [1:1000004:1] FTP Connection Attempt [**] [Priority: 0] {TCP} 10.35.209.128:36600 -> 10.35.209.233:2
1
03/19-13:42:00.774637  [**] [1:1000004:1] FTP Connection Attempt [**] [Priority: 0] {TCP} 10.35.209.128:36600 -> 10.35.209.233:2
1
03/19-13:42:00.818742  [**] [1:1000004:1] FTP Connection Attempt [**] [Priority: 0] {TCP} 10.35.209.128:36600 -> 10.35.209.233:2
```

✓ Telnet Connection Detection

```
kiran@kiran: /etc/snort/rules                    kiran@kiran: /etc/snort
00:27ff:feb6:23e1
03/19-13:45:20.650266  [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] {IPV6-ICMP} fe80::a00:27ff:feb6:23e1 -> fe80::fc
e1:81ff:fee1:4cd0
03/19-13:45:21.279097  [**] [1:1000008:1] SYN Flood Attack Detected [**] [Priority: 0] {TCP} 10.35.209.128:57640 -> 10.35.209.23
3:23
03/19-13:45:21.279097  [**] [1:1000005:1] Telnet Connection Attempt [**] [Priority: 0] {TCP} 10.35.209.128:57640 -> 10.35.209.23
3:23
03/19-13:45:21.773874  [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] {IPV6-ICMP} fe80::fce1:81ff:fee1:4cd0 -> ff02::1
:ff10:2915
03/19-13:45:21.808428  [**] [1:1000008:1] SYN Flood Attack Detected [**] [Priority: 0] {TCP} 10.35.209.128:57646 -> 10.35.209.23
3:23
03/19-13:45:21.808428  [**] [1:1000005:1] Telnet Connection Attempt [**] [Priority: 0] {TCP} 10.35.209.128:57646 -> 10.35.209.23
3:23
03/19-13:45:22.358568  [**] [1:1000008:1] SYN Flood Attack Detected [**] [Priority: 0] {TCP} 10.35.209.128:57650 -> 10.35.209.23
3:23
03/19-13:45:22.358568  [**] [1:1000005:1] Telnet Connection Attempt [**] [Priority: 0] {TCP} 10.35.209.128:57650 -> 10.35.209.23
3:23
03/19-13:45:22.593385  [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] {IPV6-ICMP} fe80::fce1:81ff:fee1:4cd0 -> ff02::1
:ff10:2915
03/19-13:45:22.925805  [**] [1:1000008:1] SYN Flood Attack Detected [**] [Priority: 0] {TCP} 10.35.209.128:57656 -> 10.35.209.23
3:23
03/19-13:45:22.925805  [**] [1:1000005:1] Telnet Connection Attempt [**] [Priority: 0] {TCP} 10.35.209.128:57656 -> 10.35.209.23
3:23
03/19-13:45:23.484724  [**] [1:1000008:1] SYN Flood Attack Detected [**] [Priority: 0] {TCP} 10.35.209.128:50040 -> 10.35.209.23
```

✓ DNS Query Detection

```
kiran@kiran: /etc/snort/rules                    kiran@kiran: /etc/snort
kiran@kiran:/etc/snort/rules$ sudo snort -A console -q -c /etc/snort/snort.conf -i enp0s3
03/19-13:39:29.922568  [**] [1:1000006:1] DNS Query Detected [**] [Priority: 0] {UDP} 10.35.209.128:40901 -> 10.35.209.233:53
03/19-13:39:29.922640  [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] {ICMP} 10.35.209.233 -> 10.35.209.128
03/19-13:39:29.926342  [**] [1:1000006:1] DNS Query Detected [**] [Priority: 0] {UDP} 10.35.209.128:59201 -> 10.35.209.233:53
03/19-13:39:29.926385  [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] {ICMP} 10.35.209.233 -> 10.35.209.128
03/19-13:39:29.929370  [**] [1:1000006:1] DNS Query Detected [**] [Priority: 0] {UDP} 10.35.209.128:36693 -> 10.35.209.233:53
03/19-13:39:29.929410  [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] {ICMP} 10.35.209.233 -> 10.35.209.128
```

✓ SYN Flood Attack Detection

```
kiran@kiran: /etc/snort/rules                    kiran@kiran: /etc/snort
3:40911
03/19-13:47:15.851671  [**] [1:1000008:1] SYN Flood Attack Detected [**] [Priority: 0] {TCP} 10.35.209.128:48636 -> 10.35.209.23
3:5431
03/19-13:47:15.851672  [**] [1:1000008:1] SYN Flood Attack Detected [**] [Priority: 0] {TCP} 10.35.209.128:48636 -> 10.35.209.23
3:1271
03/19-13:47:15.854720  [**] [1:1000008:1] SYN Flood Attack Detected [**] [Priority: 0] {TCP} 10.35.209.128:48636 -> 10.35.209.23
3:5000
03/19-13:47:15.854722  [**] [1:1000008:1] SYN Flood Attack Detected [**] [Priority: 0] {TCP} 10.35.209.128:48636 -> 10.35.209.23
3:1700
03/19-13:47:15.854723  [**] [1:1000008:1] SYN Flood Attack Detected [**] [Priority: 0] {TCP} 10.35.209.128:48636 -> 10.35.209.23
3:1301
03/19-13:47:15.854724  [**] [1:1000008:1] SYN Flood Attack Detected [**] [Priority: 0] {TCP} 10.35.209.128:48636 -> 10.35.209.23
3:6543
03/19-13:47:15.854725  [**] [1:1000008:1] SYN Flood Attack Detected [**] [Priority: 0] {TCP} 10.35.209.128:48636 -> 10.35.209.23
3:5226
03/19-13:47:15.854726  [**] [1:1000008:1] SYN Flood Attack Detected [**] [Priority: 0] {TCP} 10.35.209.128:48636 -> 10.35.209.23
3:3017
03/19-13:47:15.854726  [**] [1:1000008:1] SYN Flood Attack Detected [**] [Priority: 0] {TCP} 10.35.209.128:48636 -> 10.35.209.23
```

- ✓ **Nmap Scan Detection**

```
kiran@kiran:/etc/snort/rules$ sudo snort -A console -q -c /etc/snort/snort.conf -i enp0s3
03/19-13:57:53.816636  [**] [1:1000009:1] Nmap Scan Detected [**] [Priority: 0] {TCP} 10.35.209.128:45097 -> 10.35.209.233:8888
03/19-13:57:54.042833  [**] [1:1418:11] SNMP request tcp [**] [Classification: Attempted Information Leak] [Priority: 2] {TCP} 1
0.35.209.128:45097 -> 10.35.209.233:161
03/19-13:57:54.194403  [**] [1:1421:11] SNMP AgentX/tcp request [**] [Classification: Attempted Information Leak] [Priority: 2]
{TCP} 10.35.209.128:45097 -> 10.35.209.233:705
03/19-13:58:08.240456  [**] [1:1228:7] SCAN nmap XMAS [**] [Classification: Attempted Information Leak] [Priority: 2] {TCP} 10.3
5.209.128:51510 -> 10.35.209.233:1
03/19-13:58:08.893796  [**] [1:1000009:1] Nmap Scan Detected [**] [Priority: 0] {TCP} 10.35.209.128:41854 -> 10.35.209.233:995
03/19-13:58:19.828105  [**] [1:1000009:1] Nmap Scan Detected [**] [Priority: 0] {TCP} 10.35.209.128:37286 -> 10.35.209.233:25
03/19-13:58:50.234400  [**] [1:1000009:1] Nmap Scan Detected [**] [Priority: 0] {TCP} 10.35.209.128:63558 -> 10.35.209.233:1723
```

- ✓ **ICMP Flood Detection**

```
kiran@kiran: /etc/snort/rules                    kiran@kiran: /etc/snort
03/19-13:52:12.184474  [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] {ICMP} 10.35.209.233 -> 10.35.209.128
03/19-13:52:12.186673  [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] {ICMP} 10.35.209.128 -> 10.35.209.233
03/19-13:52:12.186693  [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] {ICMP} 10.35.209.233 -> 10.35.209.128
03/19-13:52:12.189584  [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] {ICMP} 10.35.209.128 -> 10.35.209.233
03/19-13:52:12.189605  [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] {ICMP} 10.35.209.233 -> 10.35.209.128
03/19-13:52:12.191864  [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] {ICMP} 10.35.209.128 -> 10.35.209.233
03/19-13:52:12.191885  [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] {ICMP} 10.35.209.233 -> 10.35.209.128
03/19-13:52:12.194648  [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] {ICMP} 10.35.209.128 -> 10.35.209.233
03/19-13:52:12.194681  [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] {ICMP} 10.35.209.233 -> 10.35.209.128
03/19-13:52:12.196556  [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] {ICMP} 10.35.209.128 -> 10.35.209.233
03/19-13:52:12.196578  [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] {ICMP} 10.35.209.233 -> 10.35.209.128
03/19-13:52:12.198264  [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] {ICMP} 10.35.209.128 -> 10.35.209.233
```

Conclusion

Snort is a powerful IDS tool that helps in:

- Detecting network attacks
- Monitoring suspicious traffic
- Enhancing network security

By writing custom rules, administrators can:

- Detect specific attacks
- Monitor sensitive activities
- Prevent unauthorized access